RAG Chatbot Evaluation Document

Synthetic knowledge base for testing PDF ingestion, retrieval, grounding, and answer quality.

Document ID: RAG-TEST-001

Version: 1.0

1. Company Overview

NovaTech Systems is a fictional software company founded in 2018. Its headquarters are in Ahmedabad, Gujarat, India. The company develops workflow automation, analytics, and document-processing software for small and medium-sized businesses.

NovaTech has three primary product lines: FlowDesk, InsightBoard, and DocuVault. FlowDesk handles workflow automation, InsightBoard provides business analytics, and DocuVault provides document storage and search.

2. Products

Product	Purpose	Launch Year	Monthly Price
FlowDesk	Workflow automation	2020	■799
InsightBoard	Business analytics	2021	■999
DocuVault	Document storage and search	2023	■599

3. Technical Architecture

NovaTech's reference architecture uses a React frontend, a Node.js API layer, PostgreSQL for transactional data, Redis for caching, and object storage for large files. The authentication service uses short-lived access tokens and rotating refresh tokens.

The document-search service performs text extraction, chunking, embedding generation, vector retrieval, and answer generation. Retrieved chunks are passed to the language model as context. The system is designed so that answers should be grounded in retrieved document content.

4. Performance Facts

Metric	Target
API availability	99.9% monthly
Median API latency	180 ms
95th percentile API latency	420 ms
Maximum document size	50 MB
Maximum pages per document	500
Default retrieval count	5 chunks

5. Team and Offices

Office	Team Size	Primary Responsibility
Ahmedabad	86	Engineering and product
Bengaluru	42	Platform engineering
Pune	27	Customer success
Delhi	19	Sales and partnerships

6. Security Policy

All production API endpoints must use authenticated requests unless explicitly marked public. Passwords are never stored in plaintext. Sensitive configuration values must be stored in a secret-management system rather than committed to source control.

Customer documents are encrypted at rest. Access to production data is logged. Developers receive production access only when required for their responsibilities.

7. Incident Scenario

On 14 March 2026 at 10:20 UTC, the API monitoring system reported that approximately 10% of requests were returning HTTP 500 errors. The issue affected the document-search endpoint. The engineering team discovered that a database connection-pool configuration had been changed during a deployment at 10:05 UTC. Reverting that configuration reduced the error rate to below 0.1%.

8. RAG Evaluation Rules

A grounded answer must be supported by information contained in the retrieved document context. If the document does not contain the requested fact, the chatbot should say that the information is not available rather than inventing an answer.

For evaluation, factual questions, multi-hop questions, negative questions, paraphrased questions, and out-of-document questions should all be tested.

9. Reference Facts for Evaluation

Important exact facts: NovaTech was founded in 2018; headquarters are in Ahmedabad; FlowDesk launched in 2020 and costs ■799 per month; InsightBoard launched in 2021 and costs ■999 per month; DocuVault launched in 2023 and costs ■599 per month; the default retrieval count is 5 chunks; the maximum document size is 50 MB; the maximum page count is 500; Ahmedabad has 86 employees; the 14 March 2026 incident affected the document-search endpoint; the deployment occurred at 10:05 UTC; monitoring detected the issue at 10:20 UTC; reverting the configuration reduced the error rate below 0.1%.

End of Test Document